MIL-F-50997A (MU)
8 June 1973
SUPERSEDING
MIL-F-50997(MU)
11 April 1972

MILITARY SPECIFICATION

FILLER ASSEMBLY ALTERNATIVE (FOR USE IN AMMUNITION CONTAINERS)

1. SCOPE

1.1 Scope. This specification covers cushioning material, inclosed in close fitting waterproof barrier material, for use as filler or cushioning pads.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116 - Preservation, Methods of
MIL-A-2550 - Ammunition and Special Weapons; General Specification for

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-1169 - Packaging, Packing and Marking for Shipment of Inert Ammunition Components
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWING

U. S. ARMY MUNITIONS COMMAND

8865572 - Filler Assembly Alternative

FSC: 8140

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(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bid shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS PUBLICATIONS

D641-69 - Conditioning Paperboard, Fiberboard and Paperboard Containers for Testing
D646-67 - Basis Weight of Paper and Paperboard

(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania, 19103).

3. REQUIREMENTS

3.1 Material. Material shall be fabricated as depicted on the applicable drawing and shall be such as to produce a waterproof cushioning envelope meeting the requirements specified herein.

3.2 Weight. The dry weight of the paperboard sheet shall be as specified on the applicable drawing.

3.3 Thickness. The filler assembly shall be of the caliper specified on the applicable drawing, contract or order.

3.4 Size. Unless otherwise specified in the contract or order, the filler assembly shall be furnished in accordance with the dimensions prescribed on the applicable packing box drawing.

3.5 Resistance to water. The filler assembly shall pass the "Water Resistance Test" specified in paragraph 4.4.2.

3.6 Seam strength. The filler assembly shall be capable of withstanding the "Heat Seal Seam Test" specified in 4.4.3

3.7 Fracture. The filler assembly (except filler #2) shall be capable of bending in either direction without indication of cracking, tearing, opening of seams or other defects which may impair its usefulness. Filler #2 (constructed utilizing fluted cushioning material) shall however, be bent parallel to the direction of flutes.

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3.8 Workmanship. Workmanship shall be in accordance with the best commercial practice of the industry. Finished filler assembly shall be sound and free from imperfections which might impair its usefulness.

3.9 First article inspection. This specification makes provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define terms used herein. The provisions of Specification MIL-A-2550 shall apply.

4.1.1 Submission of product. At the time the completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified within this specification.

b. Specification number and date, together with an identification and date of changes thereto.

c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.

d. Number of items in the lot.

e. Date submitted.

The certificate shall be signed by a responsible agent of the

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certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First Article Inspection (see 6.2)

4.2.1 Submission. Prior to the initiation of regular production, the contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with the provisions of 4.2.2. The sample shall consist of 25 assemblies and 10 sets of parts which have been produced by the contractor or furnished by a supplier and which have been manufactured using the same production process, procedures and equipment which will be used in fulfilling the contract. All materials, including packaging and packing shall be obtained from the same source of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract, and submit a record of this inspection with the sample, including statements of findings for materials, processes and tests. A sample containing known defects will not be submitted unless specifically authorized by the contracting officer. A first article sample, or portion thereof, as directed by the contracting officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days, or whenever a change occurs in manufacturing process, material used, drawing or specification such as to signficantly affect product uniformity, as determined by the Government.

4.2.2 Inspection to be performed. Each Filler Assembly shall be subjected by the Government to any or all of the examinations or tests specified in this specification.

4.2.3 Rejection. If any Filler Assembly fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of any component to comply with any of the stated requirements.

4.3 Inspection provisions

4.3.1 Lot formation. The term "inspection lot", as used in

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this specification, is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer in one unchanged process, in accordance with the same drawing and same drawing revision, same specification and same specification revision, and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to the process or specification, not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of sampling procedure which is being applied to determined conformance with requirements.

4.3.2 Examination. The examinations listed in the following subparagraphs shall be performed on inspection lots (see 4.2.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of major and minor defects listed, the contractor may elect to apply AQL's to individual defects using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. When this method is elected, it shall be applied to all of the major and minor defects listed in the applicable subparagraph of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspection listed shall be in accordance with 4.3.4.

4.3.2.1 Filler Assembly Alternate, Drawing 8865572 (prior to final seal)

Categories	Defects	Method of Inspection	Code No. (see 6.3)
Critical:	None defined		
Major:	None defined		

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Minor:	AQL 1.5 percent		
201.	Length of filler	Scale(CD)	01001
202.	Width of filler	Scale(CD)	01002
203.	Sheet contains improper number of flutes (as applicable)	Visual	01003
204.	Cushioning sheet damaged,missing, of incorrect design or improperly assembled (as applicable)	Visual	01004
205.	Outer sleeve material incorrect (as applicable)	Visual	01005
206.	Outer sleeve damaged (as applicable)	Visual	01006

4.3.2.2 Filler Assembly, Alternate (see drawing 8865572)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.65 percent		
101.	Heat seal missing or incomplete	Visual	02001
102.	Cut, tear or hole evident	Visual	02002
103.	Bag incorrectly constructed (inside out)	Visual	02007
Minor:	AQL 1.5 percent		
201.	Thickness	Scale(CD)	02003
202.	Location of heat seal min	Scale(CD)	02004
203.	Fit of envelope improper (contains excessive air)	Visual	02005
204.	Evidence of poor workmanship (see 3.8)	Visual	02006

4.3.3 Testing

4.3.3.1 Conditioning

4.3.3.1.1 Unless otherwise specified, all specimens shall be brought to equilibrium in a circulating atmosphere maintained at a relative humditiy and a temperature specified in ASTM-D641-49 (1965).

4.3.3.2 Weight(see 3.2), Minor Defect, Code No. 10001. Samples to be tested shall be obtained in accordance with the provisions of ASTM D646-67. Failure of any sample to meet the applicable requirement when tested as specified in 4.4.1 shall be cause for rejection of the lot.

4.3.3.3 Resistance to water (see 3.5), Major defect, Code No. 10002. Samples to be tested shall be obtained in accordance

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with the provisions of Military Specification MIL-P-116. Failure of any sample to meet the applicable requirement, when tested as specified in paragraph 4.4.2, shall be cause for rejection of the lot.

4.3.3.4 Seam strength (see 3.6) Major defect, Code No. 10003. Samples to be tested shall be as required in Military Specification MIL-P-116. Failure of any sample to meet the applicable requirement, when tested in accordance with the provisions of 4.4.3, shall be cause for rejection of the lot.

4.3.3.5 Fracture (see 3.7), Major defect, Code No. 10004. Five samples shall be selected for this test. Failure of one or more samples to meet the applicable requirement, shall be cause for the lot to be rejected.

4.3.4 Inspection equipment. The inspection equipment required to perform the inspection prescribed in this section is identified in the Examination (4.3.2) and Test Method (4.4) paragraphs herein. See 6.4 for details concerning responsibilities for inspection equipment design and approval.

4.4 Test methods and procedures

4.4.1 Weight. The weight of the specimen shall be determined in accordance with ASTM-D646-67.

4.4.2 Resistance to water. The water resistance shall be performed in accordance with the leakage test procedure specified in Military Specification MIL-P-116.

4.4.3 Seam strength. The seam strength test shall be performed in accordance with the heat sealed seam test specified in Military Specification MIL-P-116.

4.4.4 Fracture. The test specimens selected for this test shall be conditioned to a temperature of minus(-) 20°F Fahrenheit(F) for a period of 24 hours minimum. Upon completion of conditioning the specimens shall be bent around a 3 inch mandrel 90°. Examinations shall be made for evidence of tearing, opening of seams, cracking, etc. Failure of one or more specimen to meet the applicable requirements shall be cause for rejection of the lot.

5. PREPARATION FOR DELIVERY

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5.1 Packing

5.1.1 Packing shall be Level A or C, as specified. The Filler Assemblies shall be packed per MIL-STD-1169.

5.2 Marking

5.2.1 Marking shall be in accordance with MIL-STD-1169.

6. NOTES

6.1 Intended use. Filler assemblies are designed for use where water-resistant material is required and where resistance to dimensional change under compression, rather than resiliency, is important.

6.2 Ordering data. Procurement documents shall specify the following:

a. Title, number and date of this specification
b. Level of pack
c. Size of filler assembly when drawing is not specified.
d. Provisions for submission of first article samples.
e. Filler (No. as applicable)

6.3 Inspection code numbers. The five-digit code numbers assigned to the inspections here are to facilitate future data collection and analysis by the Government.

6.4 Inspection Equipment Designs. Inspection equipment designs are of two types - Government Special Inspection Equipment (SIE) designs and contractor designs. SIE designs are designated by drawing numbers (on the Equipment Lists referenced on the Equipment Tabulation) or (under the "Method of Inspection" headings in Section 4). Design responsibility for all other inspection equipment is assigned to the contractor. Equipment to be designed by the contractor is designated "CD". The contract need not furnish any design when a complete Government SIE design is provided. Unless otherwise specified, however, the contractor may submit alternate or modified contractor designs of SIE in accordance with 6.4.2 and 6.4.3 should he elect to do so.

6.4.1 SIE designs. SIE designs may consist of any of the following:

a. Detailed drawings which completely depict all information necessary for the fabrication and use of the item of inspection equipment.

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b. A source control drawing or a specification control drawing as defined in MIL-STD-100.

c. An envelope drawing, as defined in MIL-STD-100, which establishes the criteria which a detail design shall meet. When envelope drawings are specified, the contractor shall prepare designs which comply with the criteria therein.

6.4.2 Contractor designs. Contractor designs are required for all inspection equipment for which SIE design are not specified and may include commercial equipment which the contractor proposes to use. (Commercial equipment is defined as unmodified equipment which is cataloged and available for purchase by the general public.) Contractor designs shall include appropriate operating instructions, calibration procedures and maintenance procedures. Commercial equipment shall be fully described by catalog listings or other means which provide sufficient information to permit identification and evaluation by the Government and may include illustration and engineering data. Designs shall be prepared for any special fixture(s) required to be used with commercial equipment, or with SIE designs if not otherwise covered thereby (see 6.4.1c). Designs shall be of the category and form (per MIL-D-1000) specified in the Contract Data Requirements Lists (DD Form 1423). The contractor is referred to MIL-HDBK-204, "Inspection Equipment Design" for guidance. The specification number and the applicable five-digit defect code number (or other specific identifying information) form Section 4 of this specification shall be referenced on each contractor design together with the component or assembly number and revision letter to which the specific design applies.

6.4.3 Submission of designs for approval. Contractor designs shall be approved by the Government prior to fabricating or procuring the equipment. Designs shall be submitted for approval to Picatinny Arsenal, SMUPA-QA-A-P, Dover, New Jersey, 07801 in accordance with the stipulations, time frame and distribution specified in the Contract Data Requirements List (DD Form 1423) or in the contract. Partial submission of inspection equipment designs is permissible and encouraged. However, the completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will normally be accomplished within one month after receipt.

Custodian:
Army-MU

Preparing Activity:
Army-MU

Project Number: 8140-A113

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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